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REGISTERED ATTRIBUTE-BASED ENCRYPTION VIA PAIRINGS

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BTP-I Recap: IBE from Standard CDH

Prior to this semester, we implemented the Döttling-Garg (CRYPTO 2017) non-black-box architecture to prove escrow-free IBE is theoretically possible using only standard Computational Diffie-Hellman (CDH) assumptions.

Chameleon Encryption & Garbled Circuits

Instead of compiling field mathematics directly into gates, we used **decryption-as-path-verification**.

- **Chameleon Hash:** A collision-resistant hash with a trapdoor.
- **Circuit Chain:** Ciphertext contains $n+1$ garbled circuits $(\tilde{P}_0, \ldots, \tilde{P}_{n-1}, \tilde{T})$.
- **Logic:** Output of $\tilde{P}_i$ evaluates encryptions of labels for $\tilde{P}_{i+1}$.

Key Insight: Each circuit unlocks the next in a cryptographic chain reaction.

Evaluation Chain

Input for $\tilde{P}_0$

↓ unlocks ↓

Circuit $\tilde{P}_1$

↓ unlocks ↓

$\ldots \tilde{P}_{n-1}$

↓ unlocks ↓

Final PKE Message

BTP-I Post-Presentation Improvements

Following the BTP-I presentation, we introduced optimizations to mitigate the massive computational bottlenecks of the garbled chain.

1. Tree Arity Optimization

In the naive construction, a 512-bit identity requires a depth of 512 sequential circuits.

- We restructured the evaluation tree to process the identity in blocks (128-bit segments).
- 4×128 bits for a 512-bit ID significantly reduces circuit depth.

$$d = n/\log_2(a)$$

2. Label Chunking & AES Integration

We utilized the active label $\tilde{\Pi}_i$ to derive a symmetric key K, using standard AES encryption to secure the path.

$$K \leftarrow KDF(\tilde{\Pi}_i)$$
$$C_{out} = AES_K(Next\ Label)$$
$$ct = C_{out} \oplus GC\ Payload$$

Reduced wire labels from 512-bit to 128-bit.

Practical Limits of Garbled Cryptography

Despite symmetric-key and arity optimizations, our BTP-I analysis proved that non-black-box CDH constructions are fundamentally disqualified from real-world, dynamic network scenarios.

Cryptographic Operation	Native Cost	Garbled Cost
EC Scalar Mult.	≈ 1 group op	$10^6 - 10^9$ gates
Field Multiplication	≈ 1 field op	Thousands of gates
AES-128 (Ours)	HW Accelerated	3,000 - 5,000 gates

The Bandwidth Bottleneck

A single encrypted message forces the transmission of 10^6+ garbled gate entries, instantly saturating network limits and paralyzing mobile or IoT environments.

BTP-I Conclusion

Ciphertext overhead triggers massive multi-megabyte payload explosions per transmission. We must abandon boolean circuit simulations and return to native algebraic "black-box" operations.

The Evolution of Access Control

1

Traditional PKE

All-or-nothing encryption. The sender must encrypt data targeting one specific recipient.

- **Pros:** Establishes secure 1-to-1 communication without prior secret-sharing.
- **Cons:** Massive key management overhead and costly CA infrastructure.

2

Identity-Based (IBE)

Identity as Public Key. Uses a unique string (like an email) directly as the public key.

- **Pros:** Eliminates the need for CAs.
- **Cons:** Requires absolute, high trust on the PKG.
- **Cons:** PKG has all Secret Keys (Key Escrow).

3

Attribute-Based (ABE)

Policy in the Math. Embeds boolean access rules directly into the ciphertext.

- **Pros:** Provides fine-grained, Controlled Access.
- **Pros:** Enables scalable one-to-many sharing.
- **Cons:** Still inherits the PKG trust vulnerability.

Enter Registered ABE (Reg-ABE)

A modern cryptographic breakthrough (2023) that eliminates the "trusted central authority" problem, shifting power directly back to the users.

1. User Autonomy

You hold the keys. Users generate their own secret keys right on their devices. No central server ever sees or holds your private data.

2. Key Curator

A public noticeboard. The all powerful admin is replaced by a "Key Curator" a public tracker that organizes public keys, but holds absolutely zero secrets.

3. Math as the Police

Built-in alarms. Advanced math (Zero-Knowledge proofs) acts as a strict auditor. If the Curator is hacked or tries to cheat, the math immediately rejects it.

THE BIG IDEA: We no longer have to trust a central server not to read our data. The system is secured purely by mathematics, not by promises.

The Key Escrow Bottleneck

The Central PKG Vulnerability

To issue a functional secret key, the PKG uses the Master Secret Key (MSK):

$$SK_{attr} \leftarrow KeyGen(MSK, Attributes)$$

- **Single Point of Failure:** The PKG can decrypt *every* ciphertext in the network.
- **Malicious Provider:** If the cloud provider is compromised, system-wide privacy can be permanently lost.

Registered ABE

Registered ABE replaces the PKG with a non-privileged, deterministic **Key Curator**.

$$(sk_u, pk_u) \leftarrow KeyGen(crs_{local})$$

- **Client-Side Keygen:** Users compute their own secrets.
- **Trustless Aggregation:** Curator only manages public keys.
- **Post-Compromise Security:** Curator database leaks reveal zero user secrets.

Bilinear Pairings & Group Environments

A pairing map $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ where all 3 groups share the same order. It must satisfy three strict axioms:

1. Bilinearity
$e(u^a, v^b) = e(u, v)^{ab}$

2. Non-Degeneracy
$e(g_1, g_2) \neq 1_{\mathbb{G}_T}$

3. Computability
Polynomial Time

The Group Spectrum

- **Composite-Order** (Order N)
  $N = p_1 \cdot p_2 \cdot p_3$ (where p_i are primes)
- **Prime-Order** (Order p)
  - **Type I:** Symmetric ($\mathbb{G}_1 = \mathbb{G}_2$)
  - **Type II:** Asymmetric. Easy homomorphism ($\phi : \mathbb{G}_2 \to \mathbb{G}_1$)
  - **Type III:** Asymmetric.
    (Finding ϕ is computationally hard)

Legacy: Composite

- **Pro:** Natural orthogonal subgroups allow easy mathematical cancellation (blinding factors map to 1).
- **Con:** Requires massive RSA-level modulus sizes (≥ 3072-bit) to resist modern GNFS factoring attacks.
- **Con:** Exceptionally slow pairing operations and giant key sizes make it entirely impractical for mobile or IoT networks.

Modern: Type III

- **Compact:** BLS12-381 curves yield highly compressed parameters offering 128-bit security.
- **Separation:** Distinct groups ($\mathbb{G}_1 \neq \mathbb{G}_2$) prevent mix-and-match attacks between keys and ciphertexts.
- **Tight Proofs:** The inability to compute ϕ prevents structural leakage, enabling tight k-Lin reductions in the standard model.

Algebraic Foundations & Assumptions

Transitioning to prime-order pairing cryptography requires understanding the specific hardness assumptions that anchor semantic security.

Subgroup Decision Assumption

Anchors legacy composite-order cryptography.

- Group order $N = p_1 p_2 p_3$
- Adversary cannot discern if an element resides in the full group $\mathbb{G}_N$ or a constrained subgroup $\mathbb{G}_{p1}$.
- Allows independent subgroup cancellation, but requires very large (1024-2048 bit) key sizes.

k - Linear (k - Lin) Assumption

Anchors modern prime-order cryptography. Generalizes DDH ($k = 1$) and DLIN ($k = 2$).

- Given vectors in an affine subspace of dimension k
- Computationally intractable to determine if a challenge vector belongs to that specific subspace or is uniform.
- Allows for 256-bit parameter sizes.

Dual Pairing Vector Spaces (DPVS)

To simulate composite subgroup cancellations in highly efficient prime-order spaces, we **replace group elements with vectors**.

Orthogonal Bases

Setup algorithm generates two strictly orthogonal bases:

$$\mathbb{B} = (b_1, b_2, b_3, b_4) \in \mathbb{G}_1^4$$

$$\mathbb{B}^* = (b_1^*, b_2^*, b_3^*, b_4^*) \in \mathbb{G}_2^4$$

Dot-product pairing yields orthogonality:

$$e(b_i, b_j^*) = \begin{cases} g_T^{\psi} & \text{if } i = j \\ 1 & \text{if } i \neq j \end{cases}$$

Architectural Roles

- **Master Public Key & Ciphertexts** live in the span of $\mathbb{B}$ (in $\mathbb{G}_1$).
- **Secret Keys & Curator's HSK** live in the span of $\mathbb{B}^*$ (in $\mathbb{G}_2$).
- **Blinding Subspaces:** Cross-terms generated during curator aggregation are placed in specific subspaces (e.g., spanned by b_2, b_3).

Result: During decryption, orthogonality mathematically guarantees $e(b_{data}, b_{blind}^*) = 1$, canceling the blinding factors while preserving the secret payload.

Linear Secret Sharing Schemes (LSSS)

Cryptographic keys and ciphertexts are intrinsically associated with a policy construction. Raw boolean logic must be mathematically converted into matrices to enforce these access structures within pairing environments.

LSSS Construction

- **Policy Matrix:** $\ell \times n$ matrix M with mapping function ρ (rows $\rightarrow$ attributes).
- **Secret Vector:** Encryptor hides secret s inside $\vec{v} = (s, r_2, ..., r_n)^\top$.
- **Share Generation:** Multiplication $M \cdot \vec{v}$ yields the shares.
- **Reconstruction:** If attributes satisfy the structure, user derives constants c_i to solve $\sum c_i(M_i \cdot \vec{v}) = s$.

ⓘ Non-Monotone Span Programs

LSSS mathematically represents NOT-gates via orthogonal vectors, natively supporting exclusionary logic (e.g., `Director AND NOT Contractor`).

Matrix Projection Visualization

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} s \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix}$$

*Shares λ_i embed the payload. Only correct linear combinations of λ_i cancel out random factors r_x.

The Key Curator Model: Core Algorithms

1 Setup

Inputs: Security param 1^λ, max users 1^N

Outputs: Public parameters crs.

No Master Secret Key is generated.

2 KeyGen

Inputs: Parameters crs

Outputs: Secret key sk_i, Public key pk_i, and a well-formedness proof π_i.

3 Is_Valid

Inputs: crs, Public key pk_i, Proof π_i

Outputs: 1 (valid) or 0 (invalid). Prevents malicious key injection into the MPK.

4 Aggregate

Inputs: Parameters crs, State st, Public key pk_i, Attributes x_i

Outputs: Updated State st', Global Master Public Key mpk, and individualized Helper Decryption Keys hsk_i.

5 Encrypt

Inputs: Master Public Key mpk, Policy y, Message m

Outputs: Ciphertext ct.

6 Decrypt

Inputs: Secret key sk_i, Helper Key hsk_i, Ciphertext ct

Outputs: Message m (if attributes satisfy policy), else $\perp$.

Vector Commitments & Merkle Registration

Achieving $O(\log N)$ registration efficiency and trustless auditing.

The Registration Tree

Public keys are compressed into a digest via Vector Commitments in pairing group $\mathbb{G}$:

$$C = \prod_{i \in [N]} h_i^{pk_i}$$

- Commitments form leaves of a Merkle Tree.
- The **Root Hash** definitively acts as the MPK.
- Any user can request $O(\log N)$ sibling hashes to verify inclusion without trusting the Curator.

Slotted Merging

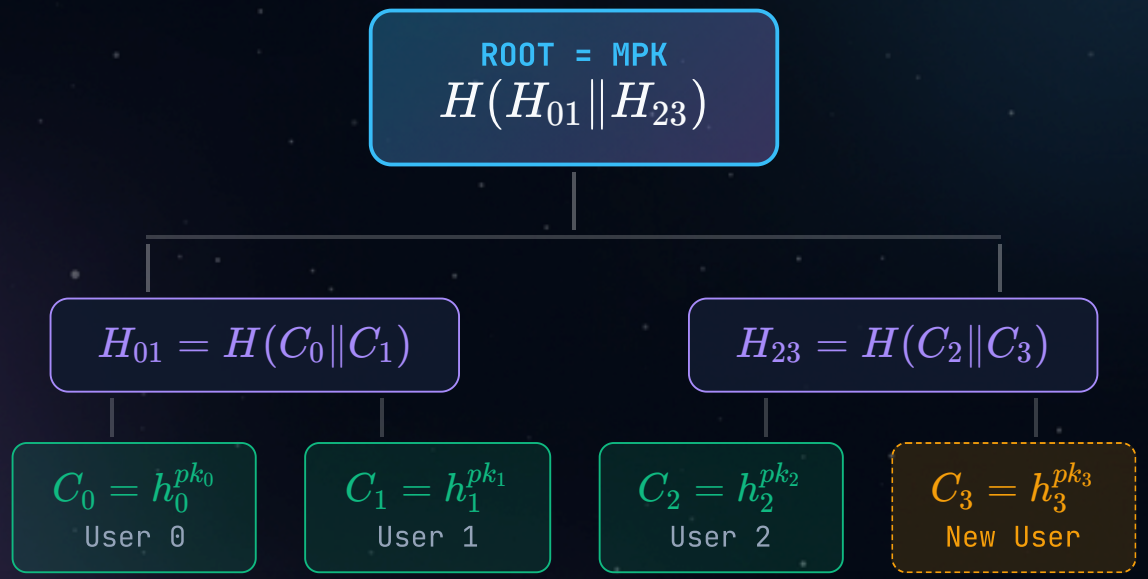
Direct dynamic joining is complex. We implement a "Powers-of-Two" mechanism over slotted capacities.

Tree size 8 | Tree size 4 | Tree 2 | Tree 1

When User 16 registers, they fill a new slot of size 1. This triggers a cascading merge of trees $1 \to 2 \to 4 \to 8 \to 16$.

Merkle Registration Tree — Visual Structure

The Curator organizes all registered public keys into a binary Merkle tree. The root hash serves as the compact Master Public Key (MPK).



Inclusion Proof: User 2 only needs sibling C_3 and hash H_{01} — just $O(\log N)$ elements — to verify their membership against the root MPK.

Dynamic Updates: When User 3 registers (dashed), only H_{23} and the Root are recomputed. All other branches remain unchanged.

Merging Keys & Filtering Noise

A high-level view of how the Curator combines keys, and how pairings filter out the resulting noise.

Step 1: Aggregate

User i samples a local secret $r_i \leftarrow \mathbb{Z}_p$ and publishes $T_i = g^{r_i}$. The Curator aggregates these purely via multiplication:

$$\hat{T} = \prod T_j = g^{\sum r_j}$$

The Challenge: Because the global Master Public Key incorporates exponents from all users, it generates cryptographic interference known as **cross-terms**.

Step 2: Decrypt

To decrypt, a user must filter out everyone else's noise. The Curator provides blinding factors $V_{j,i} = A_j^{r_i}$ inside the Helper Secret Key (hsk_i).

$$e(C, h_{N+1-i}) = e(\Lambda_i, g_2) \cdot e(h_i^{r_i}, h_{N+1-i})$$

The Solution: The bilinear pairing $e(\cdot, \cdot)$ acts as a mathematical filter, elegantly canceling out the cross-terms to isolate the user's specific slot.

Predicate Encodings ($LSSS$ Policy Checks)

Maps boolean access formulas into matrices over a finite field $\mathbb{Z}_p$. Let $x \in \mathcal{X}$ (attributes) and $y \in \mathcal{Y}$ (policy).

1. Encoding Components

Ciphertext Encoding (C_x): Mathematical attribute vector.
Key Encoding (K_y): The LSSS access matrix.
Decryption Vector ($d_{x,y}$): Computable *only* if $P(x, y) = 1$.

2. The Reconstruction

User computes $M_{x,y} d_{x,y}^\top = e_1^\top$, effectively wiping the blinding factors. If $P(x, y) = 0$, shared secret α remains perfectly statistically hidden.

COMPOSITE MATRIX

$$M_{x,y} = \begin{pmatrix} \alpha_y & 0 \\ K_y & C_x \end{pmatrix}$$

Secret Payload
α_y

Blinding Subspace
K_y, C_x

DECRYPTION FLOW

Attributes → C_x + Policy → K_y → Build $M_{x,y}$ → Solve $d_{x,y}$ → Recover α

Efficiency Optimization via Prime-Order Groups

Moving from composite subgroups to prime-order vector subspaces (via DPVS) drastically reduces overhead.

Metric	Composite-Order Group	Prime-Order Vector Space
Math Basis	$N = p_1 \cdot p_2 \cdot p_3$ (Factoring Hard)	Single Prime p (e.g. BLS12-381)
Isolation Tool	Subgroup $\mathbb{G}_{p1}$ cancels $\mathbb{G}_{p2}$	Orthogonal vector subspaces $\mathbb{V}_1 \perp \mathbb{V}_2$
Security Proof	Subgroup Decision Assumption	Decision Linear (k-Lin) Assumption
Ciphertext Size	Large (~1024-bit per element)	Small (~256-bit per element)
Master Public Key	$O(L^2) \rightarrow \approx 15\ \mathrm{MB}$	$O(L) \rightarrow \approx 7.5\ \mathrm{KB}$ (>2000x smaller)

2000x Smaller MPK

From ~15 MB down to ~7.5 KB, enabling mobile and IoT deployment scenarios previously impossible.

4x Faster Pairings

Prime-order pairings on BLS12-381 are ~4x faster than composite-order equivalents at the same security level.

Standard k-Lin Security

Security reductions under k-Lin are tighter and better understood than composite subgroup decision assumptions.

Kronecker Products & QA-NIZK Proofs

QA-NIZK Injection Defense

Quasi-Adaptive Non-Interactive Zero-Knowledge proofs appended to registered public keys prevent adversaries from inserting malformed components.

- $Verify(crs, pk_i, \pi_i) == 1$ required before curator aggregation.
- Defends against adaptive chosen-key attacks on the MPK structure.

Kronecker Encodings

LSSS access matrices are mapped onto DPVS basis networks via Kronecker Products ($\otimes$):

$$A \otimes B = \begin{pmatrix} a_{11}B & \dots & a_{1m}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \dots & a_{nm}B \end{pmatrix}$$

Full CCA-Security

By combining QA-NIZK proofs of ciphertext well-formedness with one-time signatures binding the ciphertext to the policy, the system is upgraded from CPA (passive) to **Chosen-Ciphertext Attack (CCA) resilience**, securing the payload against aggressive decryption oracle manipulation.

Simulation-Based Security Proofs

Proving security in Reg-ABE relies on the **Dual-System Argument**, which controls entropy leakage across sequential key queries.

IND-Reg-ABE Game

1. Adversary chooses messages m_0, m_1 and challenge policy y^*.
2. Challenger encrypts m_b (random b) under y^*.
3. Adversary queries $Register(x_i)$ for any attribute set where $P(x_i, y^*) = 0$ (ensuring the challenge policy cannot be satisfied).
4. Adversary guesses b'. Advantage must be negligible: $|\Pr[b' = b] - 1/2| \leq \epsilon$.

Hybrid Transition (k-Lin)

- **Game 0:** Real IND-Reg-ABE game.
- **Game 1:** Replace ciphertext randomness with DPVS blinding.
- **Game 2:** Apply k-Lin reduction.
- **Game 3:** Dual-system transition (keys become semi-functional).
- **Game 4:** Perfect information-theoretic hiding of message payload.

Registered Functional Encryption (Reg-FE)

Standard ABE decryption is strictly binary (all or nothing). Functional Encryption allows a user to evaluate a function $f(x)$ over an encrypted payload x, outputting exactly the computed result and **revealing nothing else** about the data.

Registered Inner-Product FE (Reg-IPFE)

User registers a linear function vector $\vec{y}$. Provider encrypts sensitive data vector $\vec{x}$ (e.g., financial metrics).

$$f_{\vec{y}}(\vec{x}) = \langle \vec{x}, \vec{y} \rangle = \sum_{i=1}^{n} x_i y_i$$

- **Ciphertext Size:** Weakly compact $O(n \cdot \log L)$ group elements.
- **Security:** Adaptive IND-security anchored by k-Linear assumption in standard model.
- **Application:** Decentralized statistical processing without central trust.

Registered Quadratic FE (Reg-QFE)

Quadratic Evaluation

Allows evaluation of quadratic polynomials (e.g. variances, spatial distance) over blinded arrays.

User registers matrix $F \in \mathbb{Z}_p^{n \times n}$. The ciphertext securely encodes $(\vec{x}_1, \vec{x}_2)$. The decryption sequence computes:

$$f(\vec{x}_1, \vec{x}_2) = (\vec{x}_1 \otimes \vec{x}_2) F^\top$$

Pre-Constrained Innovation (PReg-QFE)

The Problem: Computing quadratic functions directly produces non-linear terms that break the Curator's clean linear aggregation symmetry.

The Solution: Separate function definition from user registration. The master matrix defining the function space is embedded directly into the CRS during trusted setup.

Curator workload remains purely linear (simple multiplications). Non-linear complexity is enforced dynamically during decryption pairing.

Open Problems & Roadblocks

As this architecture advances, several critical challenges remain at the frontier of decentralized cryptography.

Post-Quantum Reg-ABE

Bilinear pairings on elliptic curves are fundamentally vulnerable to Shor's algorithm. Constructing efficient Reg-ABE from lattice-based assumptions (LWE) without reverting to Garbled Circuits requires novel research into lattice vector commitments.

Compactness in Reg-QFE

Current Reg-QFE suffers from a massive CRS of size $O(n^2 \cdot L^2 \cdot \log L)$. Reducing this to a linear dependency on n and L requires new dual-system algebra techniques.

Multi-Authority Extension

Current Reg-ABE assumes a single Curator. Extending to decentralized multi-authority settings — where multiple independent curators manage disjoint attribute universes — requires cross-domain consistency protocols.

Revocation Efficiency

Revoking a registered user requires Curator to recompute the aggregated MPK and issue new hsk values for all remaining users. Achieving $O(\log N)$ revocation (vs. current $O(N)$) remains an open problem.

PAIRING-BASED REGISTERED ABE

Thank You

Questions & Discussion

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